

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

WORKED SOLUTIONS

CLASS

INDEX NUMBER

CHEMISTRY

9729/02

Paper 2 Structured Questions

24 August 2017

Candidates answer on the question paper.

2 hours

Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group on all the work you hand in.

Write in dark blue or black pen.

You may use pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.

You are advised to show all working in calculations.

You are reminded of the need for good English and clear presentation in your answers.

You are reminded of the need for good handwriting.

Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets []
at the end of each question or part question.

At the end of the examination, fasten all your work
securely together.

For Examiner's Use	
1	19
2	12
3	10
4	10
5	24
Significant Figures and Units	
Handwriting and Presentation	
Total	75

This document consists of **18** printed pages.



Innova Junior College

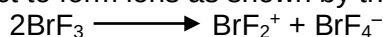
[Turn over

Answer **all** the questions in the spaces provided.

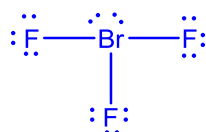
- 1 Halogens are highly reactive and form compounds with many other elements, including metals and non-metals.

(a) Fluorine reacts with bromine to form liquid bromine trifluoride, BrF_3 .

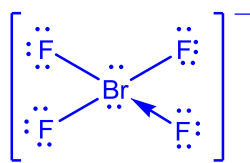
Two molecules of BrF_3 react to form ions as shown by the following equation.



- (i) Draw the structures and suggest the shape of each of these species. [3]



T-shaped



Square planar

- (ii) BrF_4^- ions are also formed when potassium fluoride, KF dissolves in liquid BrF_3 to form KBrF_4 . Explain, in terms of structure and bonding, why KBrF_4 has a high melting point. [2]

KBrF_4 has a **giant ionic structure** with **strong electrostatic forces of attraction between oppositely charged ions**, K^+ and BrF_4^- .

Large amount of energy is required to break the strong ionic bonds, thus it has a high melting point.

- (b) Magnesium bromide, MgBr_2 is a chemical compound of magnesium and bromine and is often used in the pharmaceutical industries.

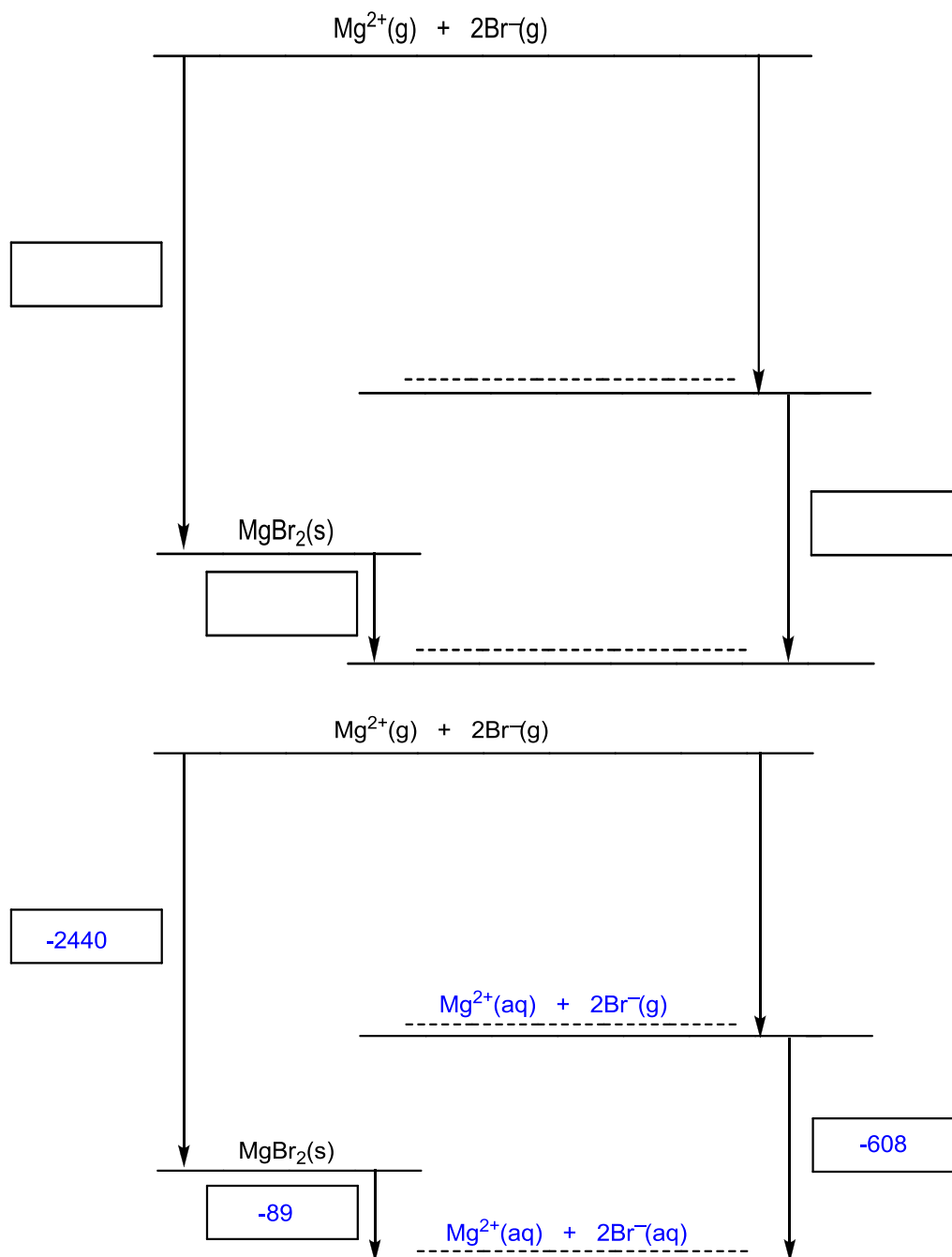
In this part of the question, you will construct an energy cycle for magnesium bromide to determine the enthalpy change of hydration of magnesium ions.

- (i) Magnesium bromide has a lattice energy of $-2440 \text{ kJ mol}^{-1}$. Define in words the *lattice energy of magnesium bromide*. [1]
Lattice energy of MgBr_2 is the **enthalpy change/ energy released** when **ONE mole of solid MgBr_2 is formed from its separate gaseous ions, $\text{Mg}^{2+}(\text{g})$ and $\text{Br}^-(\text{g})$** .

- (ii) The table below shows the enthalpy changes that are needed to determine the enthalpy change of hydration of magnesium ions.

enthalpy change	energy /kJ mol ⁻¹
lattice energy of magnesium bromide	-2440
enthalpy change of solution of magnesium bromide	-89
enthalpy change of hydration of bromide ions	-304

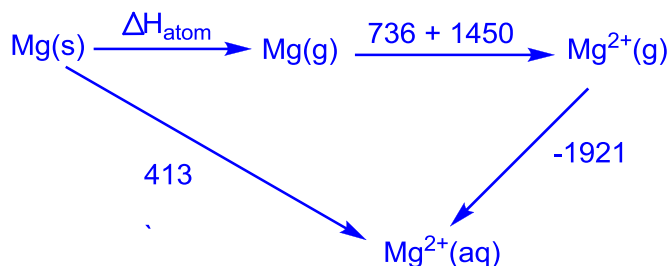
On the two dotted lines, add the species present and label the numerical values of the three arrows in the boxes provided. [2]



Species on the 2 dotted lines
Three numerical values

- (iii) Calculate the enthalpy change of hydration of magnesium ions. [1]
- Enthalpy change of hydration = $-2440 - 89 + 608 = -1921$
 $= \underline{\underline{-1920 \text{ kJ mol}^{-1}}}$

- (iv) Given that the enthalpy change of formation of $\text{Mg}^{2+}(\text{aq})$ is $+413 \text{ kJ mol}^{-1}$, use your answer in (b)(iii) as well as relevant values from the *Data Booklet* to calculate the enthalpy change of atomisation of magnesium. [1]



$$\Delta H_{\text{atomisation}} = 413 + 1921 - 736 - 1450$$

$$= +148 \text{ kJ mol}^{-1}$$

- (v) The enthalpy change of hydration of magnesium ions is more exothermic than the enthalpy change of hydration of calcium ions. Explain why. [1]

- $\Delta H_{\text{hydration}} \propto \frac{q_+}{r_+}$ or in words
- Mg^{2+} has **higher charge density than Ca^{2+}** since Mg^{2+} has a **smaller ionic size than Ca^{2+}** and hence a more exothermic $\Delta H_{\text{hydration}}$.

- (c) Nickel is a typical transition element in the d-block of the Periodic Table. Many nickel ions are able to interact with ligands to form complex ions, such as $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$.

A student dissolves nickel(II) sulfate in water. A green solution forms containing the complex ion $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$. The student then reacts separate portions of the green solution of nickel(II) sulfate as outlined below.

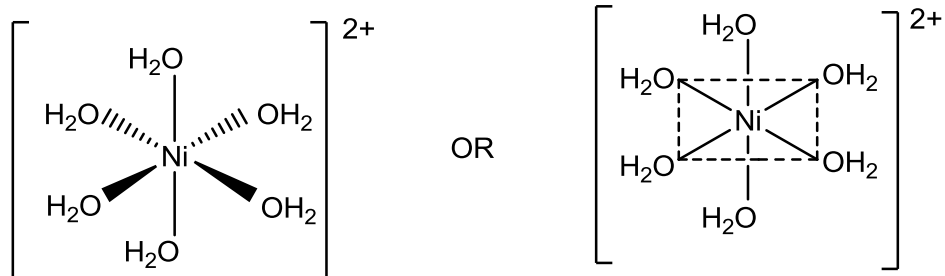
- Concentrated hydrochloric acid is added to the green solution of nickel(II) sulfate until there is no further change. The solution turns a lime-green colour and contains the four-coordinate complex ion **A**.
- Aqueous sodium hydroxide is added to the green solution of nickel(II) sulfate. A pale-green precipitate **B** forms.
- Concentrated aqueous ammonia is added to the green solution of nickel(II) sulfate until there is no further change. The solution turns a violet colour and contains the complex ion **C**. **C** has a molar mass of 160.7 g mol^{-1} .

- (i) Explain why aqueous nickel(II) sulfate is green. [2]

$\text{NiSO}_4(\text{aq})$ is coloured due to:

- the presence of **partially filled 3d-orbitals** in the Ni^{2+} metal ions.
- In the **presence of ligands, 3d orbitals split into two groups with small energy gap**
- Some of the **light energy is used to promote an electron from a d-orbital of lower energy into the unfilled/ partially filled d-orbital of higher energy.**
- Green colour seen is the **complement of the red colour absorbed** in the visible region of the spectrum.

- (ii) Draw a three dimensional diagram for the $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ ion. [1]

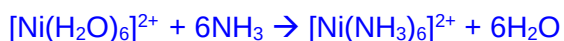


- (iii) Suggest the formulae of **A**, **B** and **C**. [3]



- (iv) What type of reaction has taken place in the formation of **C** from $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$?
Write an equation for this reaction. [2]

Ligand exchange



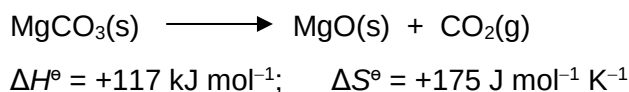
[Total: 19]

- 2 (a) In a school lab, magnesium strips can be stored in a normal container. However, barium metal has to be submerged in oil when it is stored. With reference to relevant data from the *Data Booklet*, explain the above observation. [2]

Reduction potential for Ba^{2+} (-2.92V) is more negative than Mg^{2+} (-2.38V) . Ba is more reactive OR undergoes oxidation more readily than Mg, thus has to be kept in oil to prevent it from reacting OR avoid direct contact with atmospheric O_2 .

- (b) When solid barium carbonate is heated, no carbon dioxide is detected as it is thermally stable.

However, magnesium carbonate decomposes on heating as shown by the equation below.



- (i) Explain why barium carbonate is thermally stable while magnesium carbonate is not. [2]

Ba^{2+} ion is larger and has a smaller charge density than Mg^{2+} .

Ba^{2+} ion (has a smaller polarizing power) is able to polarize/ distort the large electron cloud of the carbonate ion less effectively than Mg^{2+} , weakening the C-O bond to a smaller extent.

More energy is required to break the C – O bond in BaCO_3 than MgCO_3 .

- (ii) Account for the sign of ΔS° in this reaction. [1]

There is an increase in disorderliness due to increase in the number of moles of gaseous particles , resulting in more ways to arrange the particles in the system.

- (iii) Calculate the standard Gibbs free energy change, ΔG° , for the decomposition of magnesium carbonate. [1]

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= +117 - \left(298 \times \frac{+175}{1000} \right) = +117 - 52.15$$

$$\approx +64.9 \text{ kJ mol}^{-1}$$

- (iv) Assuming that ΔH° and ΔS° for the reaction are independent of temperature, calculate the temperature at which the decomposition reaction becomes feasible. [1]

$$\Delta G^\circ = 0 = \Delta H^\circ - T\Delta S^\circ = +117 - \left(T \times \frac{+175}{1000} \right)$$

$$T = 669\text{K} = 396^\circ\text{C}$$

At 669K, the decomposition reaction becomes feasible.

- (c) The labels of group 2 metal carbonates fell off the bottles. In an attempt to identify the compound in one of the bottles, the following experiment was carried out.

1.68 g of an insoluble metal carbonate, DCO_3 was reacted with 100 cm^3 of hydrochloric acid of concentration $0.500 \text{ mol dm}^{-3}$. The resulting solution was then made up to 250 cm^3 with distilled water. 25.0 cm^3 of this solution required 25 cm^3 of 0.04 mol dm^{-3} sodium hydroxide for titration.

- (i) Write a balanced equation for the reaction between the metal carbonate and hydrochloric acid. [1]



- (ii) Calculate the relative atomic mass of **D** and hence using the Periodic Table, identify **D**. [4]

Amount of sodium hydroxide that was used for the titration

$$= (25/1000) \times 0.04$$

$$= \underline{1.00 \times 10^{-3}} \text{ mol (3sf)}$$

$$\text{Amount of HCl present originally} = \frac{100}{1000} \times 0.500 = 0.05000 \text{ mol}$$

Amount of HCl in 25.0 cm^3 after reaction with DCO_3

$$= 1.00 \times 10^{-3} \text{ mol}$$

$$\text{Amount of unreacted HCl in } \underline{250 \text{ cm}^3} = 1.00 \times 10^{-3} \times \frac{250}{25.0}$$

$$= 1.00 \times 10^{-2} \text{ mol}$$

Amount of HCl reacted with DCO_3

$$= \text{Amount of HCl initially} - \text{amount of unreacted HCl}$$

$$= 0.05000 - 1.00 \times 10^{-2} = 0.0400 \text{ mol}$$

$$\text{Amount of } \text{DCO}_3 \text{ reacted} = 0.0400/2 = 0.0200 \text{ mol}$$

$$M_r \text{ of } \text{DCO}_3 = 1.68 / 0.0200 = 84$$

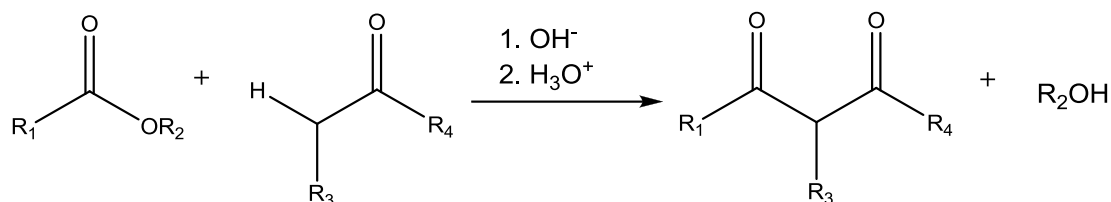
$$A_r \text{ of } \text{D} = 84 - 12.0 - 3 \times 16.0 = 24.0 \text{ (must be 1d.p.)}$$

$$\text{D} = \text{Mg}$$

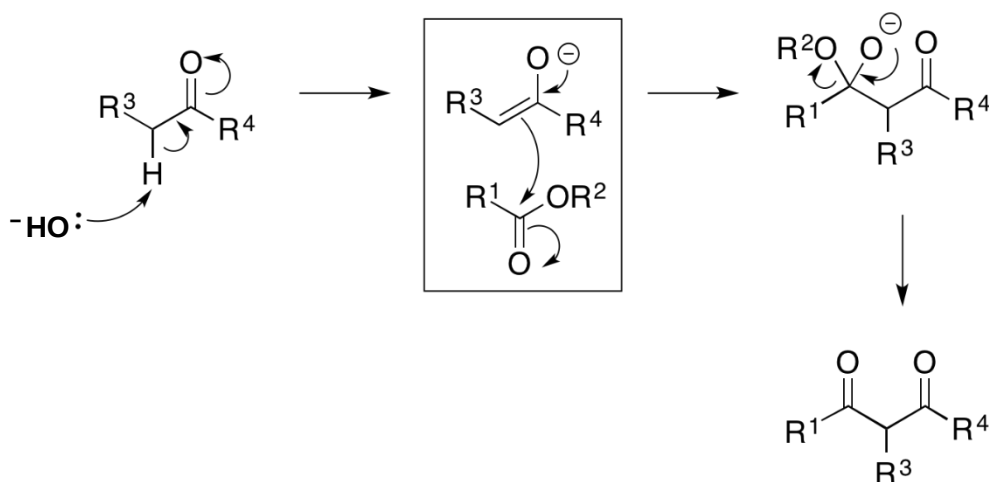
[Total: 12]

- 3 Claisen condensation is a carbon-carbon bond forming reaction between 2 esters or an ester and a carbonyl compound in the presence of strong base to form β -ketoesters or diketones respectively.

The following is the general equation for the reaction between an ester and a ketone.



The mechanism of the Claisen condensation reaction is given below.

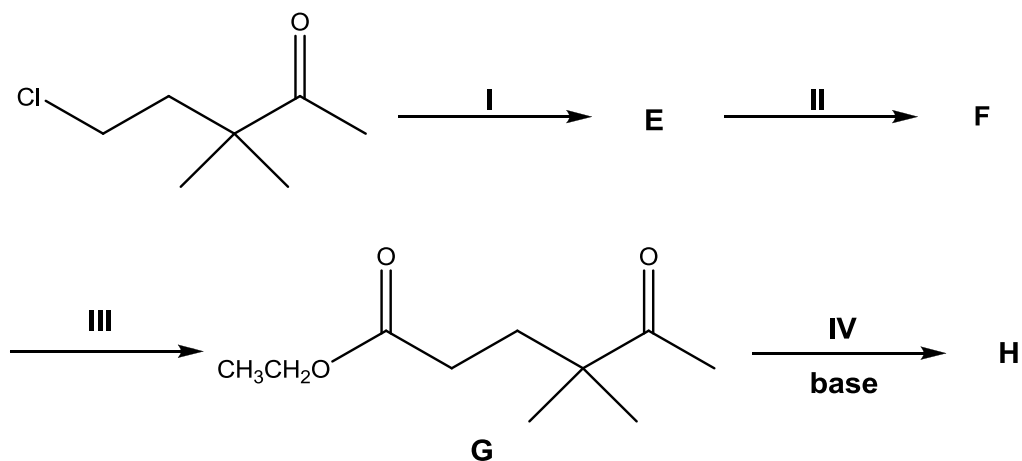


- (a) Name the type of reaction for the step shown in the box above.

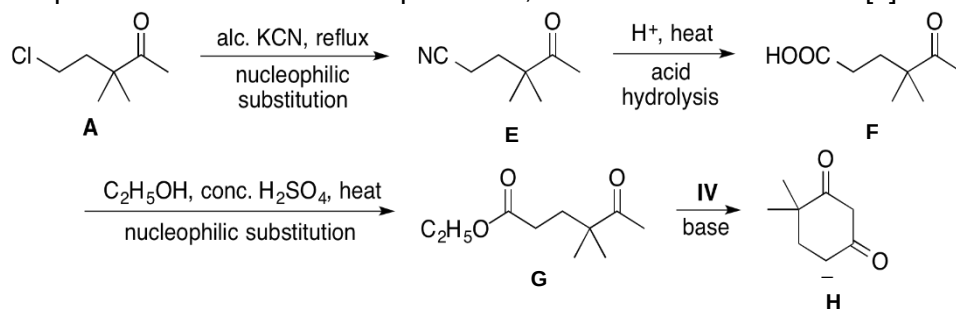
[1]

Nucleophilic addition

The Claisen condensation reaction can be used in step **IV** of the following synthesis.

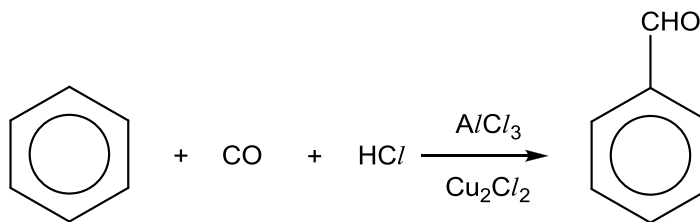


- (b) (i) Suggest reagents and conditions for steps I to III. [3]
 Step I : **ethanolic KCN, heat with reflux**
 Step II : **HCl (aq) or H₂SO₄ (aq), heat**
 Step III : **C₂H₅OH, trace amount of conc H₂SO₄, heat with reflux**
- (ii) State the type of reactions for step I to III [3]
 Step I: **Nucleophilic substitution**
 Step II: **Acidic hydrolysis**
 Step III: **Condensation or Nucleophilic acyl substitution**
- (iii) Propose the structures of compounds **E**, **F** and **H**. [3]



[Total: 10m]

- 4 (a) The Gattermann-Koch reaction, named after the German chemists Ludwig Gattermann and Julius Arnold Koch in organic chemistry, refers to a Friedel-Crafts acylation reaction in which carbon monoxide and hydrochloric acid are used in-situ with Friedel-Crafts catalyst, namely $AlCl_3$. The reaction involves an acylium ion as an electrophile and tetrachloroaluminate ion, $AlCl_4^-$. An example is provided below.



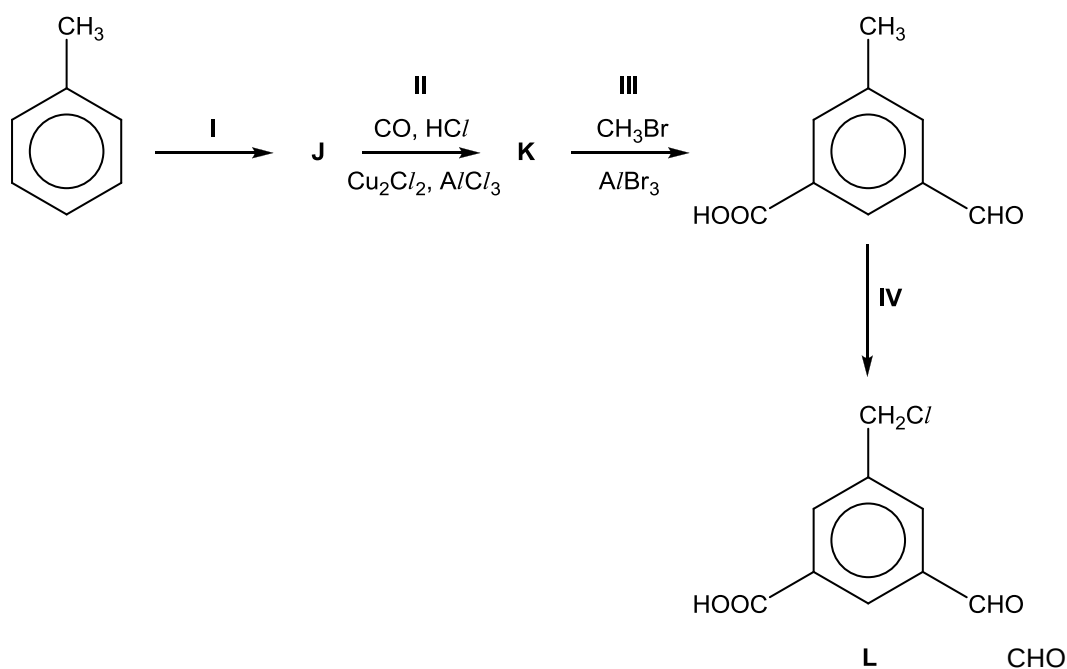
- (i) Give the formula of the acylium ion in this reaction.

[1]



- (ii) The synthesis of compound **L** involves the Gattermann-Koch reaction in Step II.

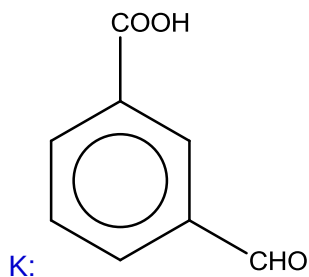
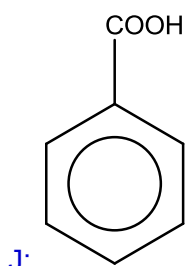
Complete the reaction scheme below by providing the appropriate reagents and conditions for Step I and IV and give the structural formula for **J** and **K**.



[4]

Step I: KMnO_4 , dil H_2SO_4 , heat with reflux

Step IV: limited Cl_2 (g) or limited Cl_2 in CCl_4 , UV light



- (b) Three non-cyclic organic compounds, **M**, **N** and **P** each have the same empirical formula CH_2O .

The number of carbon atoms in their molecules are shown in the table below.

compound	number of C atoms
M	2
N	3
P	3

All the carbon atoms are bonded directly to one another in **M** and in **N** but not in **P**.

M and **N** each give a brisk effervescence with $\text{Na}_2\text{CO}_3(\text{aq})$ but not **P**.

P does not give a silver mirror when treated with Tollens' reagent.

- (i) Draw the structural formula of **M**.

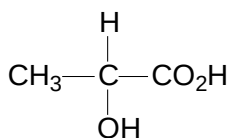
[1]



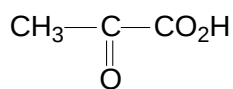
- (ii) When **N** is heated under reflux with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, the product, **Q**, gives a orange precipitate with 2,4-dinitrophenylhydrazine.

Draw the structural formulae of **N** and **Q**.

[2]



Compound **N**

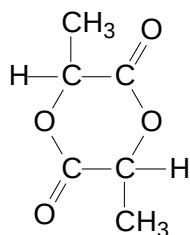


Compound **Q**

- (iii) When **N** is warmed with concentrated sulfuric acid, compound **R** is formed. **R** has the molecular formula $C_6H_8O_4$.

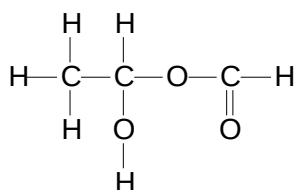
Draw the structural formula of **R**.

[1]



- (iv) **P** is optically active. Draw the displayed formula of **P**.

[1]

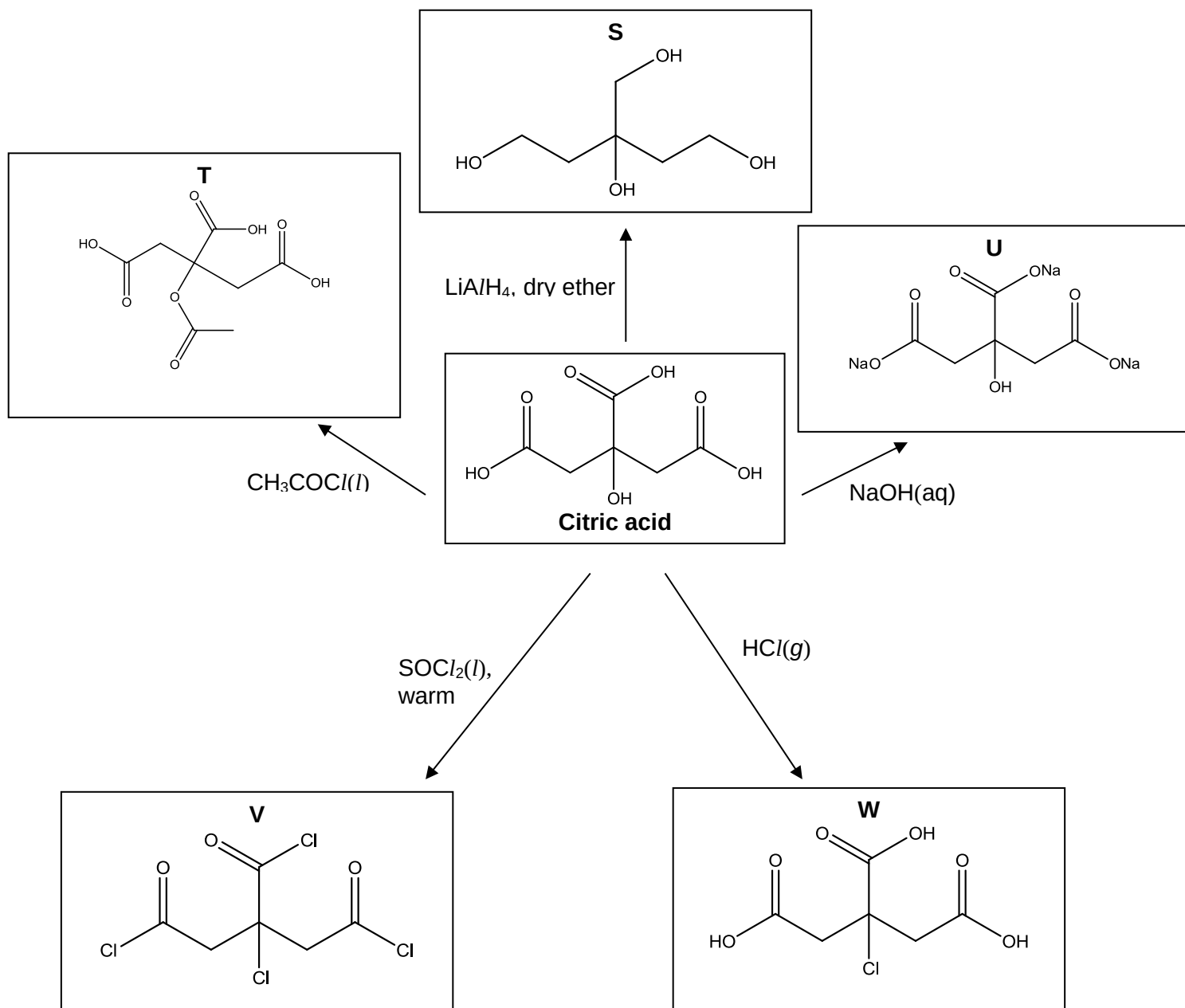


[Total: 10]

- 5 Citric acid, $C_6H_8O_7$ is produced commercially by the fermentation of sugars. Citric acid is used in the production of beverages and foods; in detergents; and in cosmetics and pharmaceuticals.

Fruits such as oranges, lemons and strawberries also contain citric acid.

(a) Citric acid can undergo the following reactions:



Suggest structures for the organic compounds **S**, **T**, **U**, **V** and **W**.

[5]

- (b) As citric acid is a user-friendly, inexpensive, water-soluble crystalline solid, it is often used for finding the concentration of alkalis e.g. sodium hydroxide and potassium hydroxide.

Citric acid is a tribasic acid with the following pK_a values:

$$pK_1 = 3.14, pK_2 = 4.75, pK_3 = 6.40$$

- (i) Explain why citric acid is soluble in water.
Citric acid has (3 **COOH groups**) which can form **hydrogen bonds** with water molecules.

[1]

- (ii) The pK_a value of ethanoic acid, CH_3COOH is 4.76.

Suggest why the pK_1 value of citric acid is lower than the pK_a value of ethanoic acid.

The anion formed by citric acid contains **two electron withdrawing –COOH group** which **disperses the negative charge** on the oxygen atom of the anion.

Thus, the anion formed by citric acid is **more stable than the ethanoate anion** and hence citric acid is **stronger**/ has a lower pK_a value.

OR

The anion formed by citric acid is further **stabilised by intramolecular hydrogen bonding** with the neighbouring COOH groups.

Thus, the anion formed by citric acid is **more stable than the ethanoate anion** and hence citric acid is **stronger**/ has a lower pK_a value .

[2]

- (iii) Calculate the pH of a solution that is a mixture of equal volumes of 0.100 mol dm^{-3} of citric acid and 0.060 mol dm^{-3} of sodium hydroxide.

A buffer solution is formed.

New initial concentration of citric acid = $0.100 / 2 = 0.050 \text{ mol dm}^{-3}$

New initial concentration of NaOH = $0.060 / 2 = 0.030 \text{ mol dm}^{-3}$



After reaction

Final concentration of citric acid salt, $\text{C}_6\text{H}_7\text{O}_7^- = 0.030 \text{ mol dm}^{-3}$

Final concentration of citric acid = $0.050 - 0.030 = 0.020 \text{ mol dm}^{-3}$

$$\begin{aligned} \text{pH of solution} &= pK_1 + \log_{10} \frac{[\text{C}_6\text{H}_7\text{O}_7^-]}{[\text{C}_6\text{H}_8\text{O}_7]} \\ &= 3.14 + \log_{10} \left[\frac{0.030}{0.020} \right] = 3.316 = \mathbf{3.32} \end{aligned}$$

[2]

- (c) The neutralisation equivalent (N.E.) of a carboxylic acid is defined as the mass in g of the acid required to neutralise one mole of a strong alkali such as sodium hydroxide. Each carboxylic acid has a unique N.E. value.

N.E. values can be used to determine the identity of a carboxylic acid.

A student carried out a titration experiment between a sample of citric acid and sodium hydroxide solution. 3.68 g of the citric acid was dissolved in 500 cm³ of water in a volumetric flask. It was found that 5.00 cm³ of the citric acid solution needed 5.29 g of 4.43 g dm⁻³ sodium hydroxide solution for complete reaction.

- (i) Calculate the mass, in g, of the citric acid used in the 5 cm³ sample.

$$\text{Mass of citric acid} = 3.68 \times \frac{5.00}{500} = 0.0368 \text{ g}$$

[1]

- (ii) Calculate the amount, in moles, of sodium hydroxide that reacted with the 5 cm³ of citric acid.

The density of the sodium hydroxide solution is 1.01 g cm⁻³.

$$\text{Concentration of NaOH} = \frac{4.43}{40} = 0.11075 \text{ mol dm}^{-3}$$

$$\text{Volume of NaOH used} = 5.29/1.01 = 5.2376 \text{ cm}^3$$

$$\begin{aligned} \text{Moles of NaOH used} &= \frac{5.2376}{1000} \times 0.11075 = 5.801 \times 10^{-4} \text{ mol (4 s.f.)} \\ &= 5.80 \times 10^{-4} \text{ mol (3 s.f.)} \end{aligned}$$

[2]

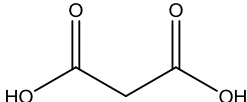
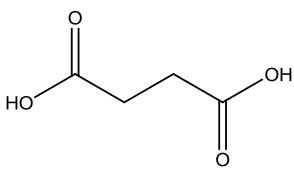
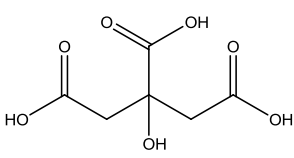
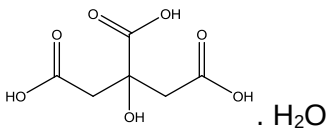
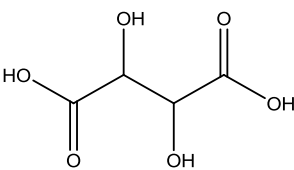
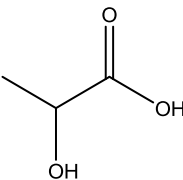
- (iii) Calculate the neutralisation equivalent (N.E.) for this sample of citric acid.

$$\text{Neutralisation Equivalent} = \frac{0.0368}{5.801 \times 10^{-4}} = 63.4$$

[1]

(d) Table 5.1 gives the neutralisation equivalent (N.E.) values for some organic acids.

Table 5.1

Acid name	Structure	Molar mass/ g mol ⁻¹	N.E. / g
Malonic acid		104.0	52.0
Succinic acid		118.0	59.0
Anhydrous citric acid		192.0	64.0
Citric acid monohydrate		210.0	70.0
Tartaric acid		150.0	<i>m</i>
Lactic acid		90.0	90.0

- (i) Suggest whether the student's sample of citric acid is a hydrated sample or anhydrous sample. Support your answer with appropriate evidence.

Anhydrous citric acid. The calculated value of N.E. is 63.4 and this is close to the theoretical value of 64.0.

[2]

- (ii) Determine the value of *m* in Table 5.1.

$$\text{N.E. for tartaric acid} = \frac{150.0}{2} = 75.0$$

[1]

- (e) Another student determines the N.E. of an unknown aliphatic acid to be 82.0 g.

The unknown aliphatic acid has the following formula, C_xH_y(COOH)_n where *x* and *y* are integers and *n* = 1, 2 or 3.

Determine the molecular formula of the unknown aliphatic acid.

The molar mass of the acid is either:

82.0 g mol⁻¹ for a monobasic acid, $n = 1$

82.0 x 2 = 164.0 g mol⁻¹ for a dibasic acid, $n = 2$

82.0 x 3 = 246.0 g mol⁻¹ for a tribasic acid, $n = 3$

The molar mass of a -COOH group = 12.0 + 2 x 16.0 + 1.0 = 45.0 g mol⁻¹

The molar mass of a -CH₂- group = 12.0 + 2.0 = 14.0 g mol⁻¹

It is a tribasic acid as **246.0 - (3 x 45.0) - (14.0 x 7) = 13.0 g mol⁻¹** (equivalent to 1C and 1 H)

The tribasic acid consists of **3 -COOH groups, 7 -CH₂- groups and one C** and one H atom. It is possible to form a molecule using these groups.

The molecular formula is **C₁₁H₁₈O₆** or **C₈H₁₅(COOH)₃**

Marking guidelines

- 1 mark for calculating the molar mass of the tribasic acid
- 1 mark for explaining why it has to be the tribasic acid using the calculated molar mass
- 1 mark for giving the correct molecular formula
- Working must be shown for the mark to be obtained. No marks to be given if the correct answer is given without any working.

Additional note (not marking points):

It cannot be a monobasic acid as $85.0 = 12.0 + 45.0 + 2 \times 14.0$

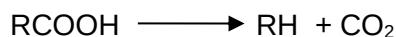
There cannot be a structure for 1 -COOH group and 2 -CH₂- groups and 1 C atom. (Molecular formula C₄H₅O₂)

It cannot be a dibasic acid as $164.0 = 2 \times 45.0 + 5 \times 14.0 + 4.0$

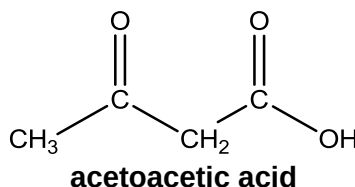
There cannot be a structure for 2 -COOH groups and 5 -CH₂- groups and 4 H atoms. (Molecular formula C₇H₁₆O₄)

[3]

- (f) The loss of carbon dioxide from a carboxylic acid is called decarboxylation.

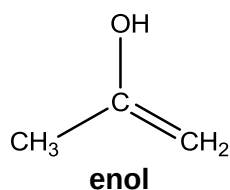


Certain types of carboxylic acids are readily decarboxylated. Beta-keto acids such as acetoacetic acid readily decarboxylate at room temperature in an acidic solution.



The decarboxylation of acetoacetic acid occurs via a two-step mechanism.

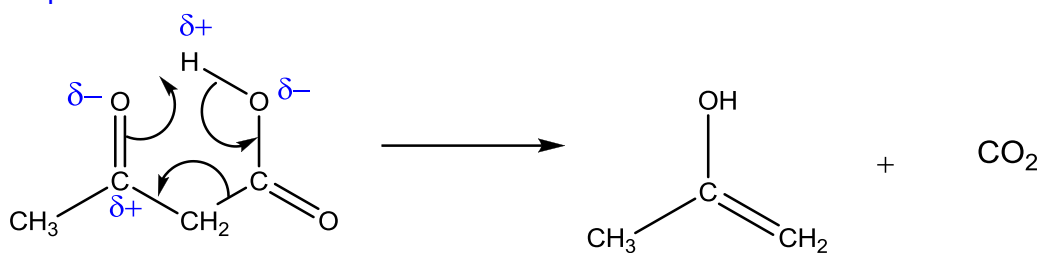
- the H atom from the COOH group is transferred to the carbonyl oxygen atom via a cyclic transition state to form CO_2 and the following compound, an enol,



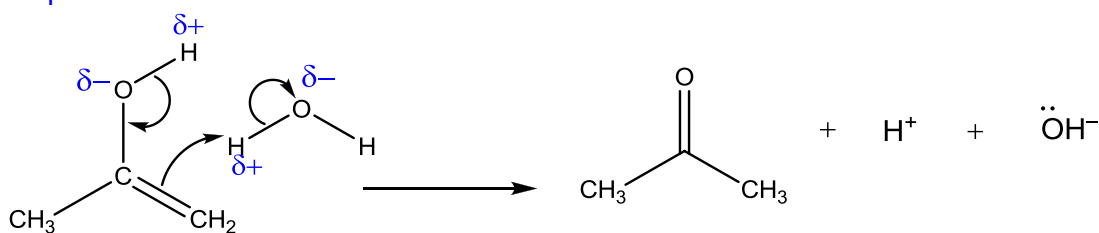
- the enol then extracts a H atom from a H_2O molecule to form a ketone and H^+ and OH^- ions.

Suggest the mechanism for this reaction. Show any relevant lone pairs, dipoles and charges, and indicate the movement of electron pairs with curly arrows.

Step 1



Step 2



[4]

[Total: 24]